**1) a)**

$$\mathbf{r} = \begin{pmatrix} 2 \\ -7 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 9 \\ -4 \end{pmatrix}$$

**1) b)**

$$\mathbf{r} = \begin{pmatrix} 3 \\ 8 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ -2 \\ 7 \end{pmatrix}$$

**2) a)**

$$\mathbf{d} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$

or just $\mathbf{r} = \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$

**2) b)**

$$\mathbf{d} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$

**2) c)**

$$\mathbf{d} = \begin{pmatrix} 3 \\ -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$

**2) d)**

$$\mathbf{d} = \begin{pmatrix} 9 \\ -3 \\ -8 \end{pmatrix} - \begin{pmatrix} 9 \\ -8 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 9 \\ -8 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 9 \\ -3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}$

**2) e)**

$$\mathbf{d} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}$

or divide the direction vector through by $-5$ to get $\mathbf{r} = \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ or

$$\mathbf{r} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

**2) f)**

$\mathbf{d} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} - \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} = \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}$

$\mathbf{r} = \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}$

or $\mathbf{r} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}$

or divide the direction vector through by $-2$ to get $\mathbf{r} = \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 7 \\ 0 \end{pmatrix}$ or

$\mathbf{r} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 7 \\ 0 \end{pmatrix}$

**3)**

$\lambda = 2$:

$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix}$

$\lambda = -2$:

$\overrightarrow{OB} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} - 2 \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \\ 0 \end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 7 \\ -7 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ -12 \\ -4 \end{pmatrix}$

$\left|\overrightarrow{AB}\right| = \sqrt{8^2 + (-12)^2 + (-4)^2} = \sqrt{224} = 4\sqrt{14}$ units

**4)**

$6 = 5 - 3\lambda \Rightarrow \lambda = -\frac{1}{3}$

$-4 = -2 + 6\lambda \Rightarrow \lambda = -\frac{1}{3}$

$-12 = -3 + 27\lambda \Rightarrow \lambda = -\frac{1}{3}$

Consistent values of $\lambda$

$\therefore (6, -4, -12)$ lies on the line

**5)**

$-12 = -2 + 4\lambda \Rightarrow \lambda = -\frac{5}{2}$

$\frac{31}{2} = 13 - \lambda \Rightarrow \lambda = -\frac{5}{2}$

$-\frac{27}{2} = -6 + 7\lambda \Rightarrow \lambda = -\frac{15}{4}$

Inconsistent values of $\lambda$

$\therefore \left(-12, \frac{31}{2}, -\frac{47}{2}\right)$ does not lie on the line

**6)**

$-2 = 7.5 + \lambda \Rightarrow \lambda = -\frac{19}{2}$

$16 = -3 - 2\lambda \Rightarrow \lambda = -\frac{19}{2}$

$-40 = 2.5 + 5\lambda \Rightarrow \lambda = -\frac{17}{2}$

Inconsistent values of $\lambda$

$\therefore (-2, 16, -40)$ does not lie on the line

**7)**

$5 = 2 + \lambda \Rightarrow \lambda = 3$

$-10 = -4 - 2\lambda \Rightarrow \lambda = 3$

$2 = k + (3) \times 3 \Rightarrow k = -7$

**8)**

$-3 = 4 + k\lambda$ (1)

$-83 = -3 + 4\lambda \Rightarrow \lambda = -20$

$-98 = 2 + 5\lambda \Rightarrow \lambda = -20$

Substitute $\lambda = -20$ into (1):

$-3 = 4 + k(-20) \Rightarrow k = \frac{7}{20}$

**9)**

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -5 \\ -9 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} = \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}$

So the line passing through $A$ and $B$ has equation $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}$

Check if $C$ lies on the line:

$\begin{pmatrix} 16 \\ 19 \\ -15 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}$

$16 = 1 - 6\lambda \Rightarrow \lambda = -\frac{5}{2}$

$19 = -1 - 8\lambda \Rightarrow \lambda = -\frac{5}{2}$

$-15 = -5 + 4\lambda \Rightarrow \lambda = -\frac{5}{2}$

Consistent values of $\lambda$

$\therefore A$, $B$ and $C$ are collinear (they lie on the same line)

**10)**

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -8 \\ 18 \\ -10 \end{pmatrix} - \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} = \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}$

So the line passing through $A$ and $B$ has equation $\mathbf{r} = \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} + \lambda \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}$

Check if $C$ lies on the line:

$\begin{pmatrix} 4 \\ 2 \\ -11 \end{pmatrix} = \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} + \lambda \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}$

$4 = 40 - 48\lambda \Rightarrow \lambda = \frac{3}{4}$

$2 = -46 + 64\lambda \Rightarrow \lambda = \frac{3}{4}$

$-11 = -18 + 8\lambda \Rightarrow \lambda = \frac{7}{8}$

Inconsistent values of $\lambda$

$\therefore A$, $B$ and $C$ are not collinear (they do not lie on the same line)

**11)**

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 11 \\ 27 \end{pmatrix} - \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} = \begin{pmatrix} -8 \\ 11-p \\ 27-p \end{pmatrix}$

So the line passing through $A$ and $B$ has equation $\mathbf{r} = \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 11-p \\ 27-p \end{pmatrix}$

$C$ lies on the line:

$\begin{pmatrix} 17 \\ -16 \\ q \end{pmatrix} = \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 11-p \\ 27-p \end{pmatrix}$

$17 = 7 - 8\lambda \Rightarrow \lambda = -\frac{5}{4}$

$-16 = p - \frac{5}{4}(11-p) \Rightarrow -64 = 4p - 55 + 5p \Rightarrow 9p = -9 \Rightarrow p = -1$

$q = -1 - \frac{5}{4}\big(27 - (-1)\big) = -36$

$\therefore p = -1$ and $q = -36$

**12)**

$-2 = k(0.5) \Rightarrow k = -4$

$8 = k(-2) \Rightarrow k = -4$

$13.5 = k(-3.375) \Rightarrow k = -4$

$\therefore \begin{pmatrix} -2 \\ 8 \\ 13.5 \end{pmatrix} = -4 \begin{pmatrix} 0.5 \\ -2 \\ -3.375 \end{pmatrix}$

Check if $(7, 8, -9)$ lies on $l_1$:

$7 = 5 - 2\lambda \Rightarrow \lambda = -1$

$8 = -2 + 8\lambda \Rightarrow \lambda = \frac{5}{4}$

$-9 = -1 + 13.5\lambda \Rightarrow \lambda = -\frac{16}{27}$

Inconsistent values of $\lambda$, so the lines are not coincident

So the lines are parallel

**13)**

$2 + 2\lambda = 4 + 3\mu \Rightarrow 2\lambda - 3\mu = 2$ (1)

$3 + \lambda = -5 + 2\mu \Rightarrow \lambda - 2\mu = -8$ (2)

$-3 + 6\lambda = 3 + 9\mu \Rightarrow 6\lambda - 9\mu = 6$ (3)

Solve (1) and (2): $\lambda = 28$ and $\mu = 18$

Check in (3):

$6(28) - 9(18) = 6$ as required

So the lines intersect.

Substitute $\lambda = 28$ into $l_1$:

The lines intersect at $(58, 31, 165)$

**14)**

$4 - 3\lambda = 6 + 8\mu \Rightarrow -3\lambda - 8\mu = 2$ (1)

$-1 + 2\lambda = -2 - \mu \Rightarrow 2\lambda + \mu = -1$ (2)

$5 + 10\lambda = 1 + 8\mu \Rightarrow 10\lambda - 8\mu = -4$ (3)

Solve (1) and (2): $\lambda = -\frac{6}{13}$ and $\mu = -\frac{1}{13}$

Check in (3):

$10\left(-\frac{6}{13}\right) - 8\left(-\frac{1}{13}\right) = -4$ as required

So the lines intersect.

Substitute $\lambda = -\frac{6}{13}$ into $l_1$:

The lines intersect at $\left(\frac{70}{13}, -\frac{25}{13}, \frac{5}{13}\right)$

**15) a)**

$6 = k(-4) \Rightarrow k = -\frac{3}{2}$

$-9 = k(6) \Rightarrow k = -\frac{3}{2}$

$16 = k\left(-\frac{32}{3}\right) \Rightarrow k = -\frac{3}{2}$

$\therefore \begin{pmatrix} 6 \\ -9 \\ 16 \end{pmatrix} = -\frac{3}{2}\begin{pmatrix} -4 \\ 6 \\ -\frac{32}{3} \end{pmatrix}$

Check if $(-4, 3, -8)$ lies on $l_1$:

$-4 = 1 + 6\lambda \Rightarrow \lambda = -\frac{5}{6}$

$3 = 9 - 9\lambda \Rightarrow \lambda = \frac{2}{3}$

$8 = -2 + 16\lambda \Rightarrow \lambda = \frac{5}{8}$

Inconsistent values of $\lambda$, so the lines are not coincident

So the lines are parallel

**15) b)**

$3 = k(-4.5) \Rightarrow k = -\frac{2}{3}$

$-8 = k(11.5) \Rightarrow k = -\frac{16}{23}$

$5 = k(-7.5) \Rightarrow k = -\frac{2}{3}$

Inconsistent values of $k$, so the lines are not parallel (or coincident).

$-5 + 3\lambda = 1 - 4.5\mu \Rightarrow 3\lambda + 4.5\mu = 6$ (1)

$-2 - 8\lambda = 1 + 11.5\mu \Rightarrow -8\lambda - 11.5\mu = 3$ (2)

$7 + 5\lambda = -7 - 7.5\mu \Rightarrow 5\lambda + 7.5\mu = -14$ (3)

Solve (1) and (2): $\lambda = -55$ and $\mu = 38$

Check in (3):

$5(-55) + 7.5(38) = 10 \neq -14$

So the lines do not intersect.

$l_1$ and $l_2$ are skew lines

**15) c)**

$2 = k(-2) \Rightarrow k = -1$

$-1 = k(6) \Rightarrow k = -\frac{1}{6}$

$-8 = k(7) \Rightarrow k = -\frac{8}{7}$

Inconsistent values of $k$, so the lines are not parallel (or coincident).

$2 + 2\lambda = 4 - 2\mu \Rightarrow 2\lambda + 2\mu = 2$ (1)

$-9 - \lambda = 5 + 6\mu \Rightarrow -\lambda - 6\mu = 14$ (2)

$3 - 8\lambda = -8 + 7\mu \Rightarrow -8\lambda - 7\mu = -11$ (3)

Solve (1) and (2): $\lambda = 4$ and $\mu = -3$

Check in (3):

$-8(4) - 7(-3) = -11$ as required

So the lines intersect.

Substitute $\lambda = 4$ into $l_1$:

The lines intersect at $(10, -13, -29)$

**16) a)**

$2 + 3\lambda = -3 + \mu \Rightarrow 3\lambda - \mu = -5$ (1)

$-4 + k\lambda = 5 - 6\mu \Rightarrow k\lambda + 6\mu = 9$ (2)

$7 + 8\lambda = -9 + 4\mu \Rightarrow 8\lambda - 4\mu = -16$ (3)

Solve (1) and (3): $\lambda = -1$ and $\mu = 2$

Substitute into (2):

$k(-1) + 6(2) = 9 \Rightarrow k = 3$

**16) b)**

Substitute $\mu = 2$ into $l_2$:

The lines intersect at $(-1, -7, -1)$

So the position vector of the point of intersection is $\begin{pmatrix} -1 \\ -7 \\ -1 \end{pmatrix}$ or $-\mathbf{i} - 7\mathbf{j} - \mathbf{k}$

**17)**

$\begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ k \\ -1 \end{pmatrix} = 0 \Rightarrow 4 \times 6 + (-5) \times k + 3 \times (-1) = 0$

$\Rightarrow 24 - 5k - 3 = 0$

$\Rightarrow 5k = 21$

$\Rightarrow k = \frac{21}{5}$

**18)**

$\begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 24 \\ 4 \\ -9 \end{pmatrix} = 2 \times 24 + (-3) \times 4 + 4 \times (-9) = 48 - 12 - 36 = 0$

Hence $l_1$ and $l_2$ are perpendicular

**19)**

$\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 2 \times 0 + (-1) \times 1 + (-1) \times (-1) = 0 - 1 + 1 = 0$

Hence $l_1$ and $l_2$ are perpendicular

**20) a)**

$\begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = (-2) \times 3 + 3 \times (-5) + 5 \times (-1) = -6 - 15 - 5 = -26 \neq 0$

So the vectors are not perpendicular

**20) b)**

$1 - 2\lambda = -13 + 3\mu \Rightarrow 2\lambda + 3\mu = 14$ (1)

$6 + 3\lambda = 28 - 5\mu \Rightarrow 3\lambda + 5\mu = 22$ (2)

$2 + 5\lambda = 24 - \mu \Rightarrow 5\lambda + \mu = 22$ (3)

Solve (1) and (2): $\lambda = 4$ and $\mu = 2$

Check in (3):

$5(4) + (2) = 22$ as required

So the lines intersect.

Substitute $\lambda = 4$ into $l_1$:

The lines intersect at $(-7, 18, 22)$

**20) c)**

Let $l_3: \mathbf{r} = \begin{pmatrix} -7 \\ 18 \\ 22 \end{pmatrix} + \gamma \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

$l_3$ is perpendicular to both $l_1$ and $l_2$ so:

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} = 0 \Rightarrow -2a + 3b + 5c = 0$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = 0 \Rightarrow 3a - 5b - c = 0$

Let $c = 1$ (for example):

$-2a + 3b + 5 = 0 \Rightarrow -2a + 3b = -5$ (4)

$3a - 5b - 1 = 0 \Rightarrow 3a - 5b = 1$ (5)

Solve (4) and (5): $a = 22$ and $b = 13$

$\therefore l_3: \mathbf{r} = \begin{pmatrix} -7 \\ 18 \\ 22 \end{pmatrix} + \gamma \begin{pmatrix} 22 \\ 13 \\ 1 \end{pmatrix}$

**21) a)**

$\cos\theta = \frac{\mathbf{d_1} \cdot \mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{1\times3+2\times(-2)+1\times2}{\sqrt{1^2+2^2+1^2}\sqrt{3^2+(-2)^2+2^2}} = \frac{1}{\sqrt{6}\sqrt{17}} = \frac{\sqrt{102}}{102}$

$\theta = \arccos\left(\frac{\sqrt{102}}{102}\right) = 84.3°$ (1 DP)

**21) b)**

$\cos\theta = \frac{\mathbf{d_1} \cdot \mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{0\times4+5\times0+2\times3}{\sqrt{0^2+5^2+2^2}\sqrt{4^2+0^2+3^2}} = \frac{6}{\sqrt{29}\sqrt{25}} = \frac{6\sqrt{29}}{145}$

$\theta = \arccos\left(\frac{6\sqrt{29}}{145}\right) = 77.1°$ (1 DP)

**21) c)**

$\cos\theta = \frac{\mathbf{d_1} \cdot \mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{3\times(-7)+(-2)\times6+(-8)\times6}{\sqrt{3^2+(-2)^2+(-8)^2}\sqrt{(-7)^2+6^2+6^2}} = \frac{-81}{\sqrt{77}\sqrt{121}} = -\frac{81\sqrt{77}}{847}$

$\theta = \arccos\left(-\frac{81\sqrt{77}}{847}\right) = 147.051\ldots°$

Acute angle is $180° - 147.051\ldots° = 32.9°$ (1 DP)

**21) d)**

$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{4\times(-15)+(-9)\times3+2\times1}{\sqrt{4^2+(-9)^2+2^2}\sqrt{(-15)^2+3^2+1^1}} = \frac{-85}{\sqrt{101}\sqrt{235}} = -\frac{85}{\sqrt{23735}}$

$\theta = \arccos\left(-\frac{85}{\sqrt{23735}}\right) = 123.485\ldots°$

Acute angle is $180° - 123.485\ldots° = 56.5°$ (1 DP)

**22) a)**

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2\times1+(-7)\times0+6\times0}{\sqrt{2^2+(-7)^2+6^2}\sqrt{1^2+0^2+0^2}} = \frac{2}{\sqrt{89}\sqrt{1}} = \frac{2\sqrt{89}}{89}$

$\theta = \arccos\left(\frac{2\sqrt{89}}{89}\right) = 77.8°$ (1 DP)

**22) b)**

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2\times0+(-7)\times1+6\times0}{\sqrt{2^2+(-7)^2+6^2}\sqrt{0^2+1^2+0^2}} = \frac{-7}{\sqrt{89}\sqrt{1}} = -\frac{7\sqrt{89}}{89}$

$\theta = \arccos\left(-\frac{7\sqrt{89}}{89}\right) = 137.901\ldots°$

Acute angle is $180° - 137.901\ldots° = 42.1°$ (1 DP)

**22) c)**

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2\times0+(-7)\times0+6\times1}{\sqrt{2^2+(-7)^2+6^2}\sqrt{0^2+0^2+1^2}} = \frac{6}{\sqrt{89}\sqrt{1}} = \frac{6\sqrt{89}}{89}$

$\theta = \arccos\left(\frac{6\sqrt{89}}{89}\right) = 50.5°$ (1 DP)

**23)**

$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{(-2)\times10+(-12)\times1+3\times5}{\sqrt{(-2)^2+(-12)^2+3^2}\sqrt{10^2+1^2+5^2}} = \frac{-17}{\sqrt{157}\sqrt{126}} = -\frac{17\sqrt{2198}}{6594}$

$\theta = \arccos\left(-\frac{17\sqrt{2198}}{6594}\right) = 96.942\ldots°$

Obtuse angle is 96.9° (1 DP)

**24)**

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{1\times2+(-1)\times4+1\times11}{\sqrt{1^2+(-1)^2+1^2}\sqrt{2^2+4^2+11^2}} = \frac{9}{\sqrt{3}\sqrt{141}} = \frac{3\sqrt{47}}{47}$$

$$\theta = \arccos\left(\frac{3\sqrt{47}}{47}\right) = 64.049\ldots°$$

Obtuse angle is $180° - 64.049\ldots° = 116.0°$ (1 DP)

**25)**

$$\begin{pmatrix}-1 & 2 & 3\\ 1 & -1 & 2\\ 0 & 2 & 1\end{pmatrix}\begin{pmatrix}3-\lambda\\ -2+6\lambda\\ 5+2\lambda\end{pmatrix} = \begin{pmatrix}-(3-\lambda)+2(-2+6\lambda)+3(5+2\lambda)\\ (3-\lambda)-(-2+6\lambda)+2(5+2\lambda)\\ 0(3-\lambda)+2(-2+6\lambda)+(5+2\lambda)\end{pmatrix}$$

$$= \begin{pmatrix}-3+\lambda-4+12\lambda+15+6\lambda\\ 3-\lambda+2-12\lambda+10+4\lambda\\ -4+12\lambda+5+2\lambda\end{pmatrix}$$

$$= \begin{pmatrix}8+19\lambda\\ 15-9\lambda\\ 1+14\lambda\end{pmatrix} = \begin{pmatrix}8\\ 15\\ 1\end{pmatrix} + \lambda\begin{pmatrix}19\\ -9\\ 14\end{pmatrix}$$

$$\therefore l_2: r = \begin{pmatrix}8\\ 15\\ 1\end{pmatrix} + \lambda\begin{pmatrix}19\\ -9\\ 14\end{pmatrix}$$

**26) a)**

$$\begin{pmatrix}2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}7+2\lambda\\ -1+5\lambda\\ 3-6\lambda\end{pmatrix} = \begin{pmatrix}2(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda)\\ 0(7+2\lambda)+1(-1+5\lambda)+0(3-6\lambda)\\ 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda)\end{pmatrix}$$

$$= \begin{pmatrix}14+4\lambda\\ -1+5\lambda\\ 3-6\lambda\end{pmatrix} = \begin{pmatrix}14\\ -1\\ 3\end{pmatrix} + \lambda\begin{pmatrix}4\\ 5\\ -6\end{pmatrix}$$

$$\therefore l_2: \mathbf{r} = \begin{pmatrix}14\\ -1\\ 3\end{pmatrix} + \lambda\begin{pmatrix}4\\ 5\\ -6\end{pmatrix}$$

**26) b)**

$$\begin{pmatrix}3&0&0\\0&3&0\\0&0&3\end{pmatrix}\begin{pmatrix}7+2\lambda\\-1+5\lambda\\3-6\lambda\end{pmatrix}=\begin{pmatrix}3(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)+3(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)+0(-1+5\lambda)+3(3-6\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}21+6\lambda\\-3+15\lambda\\9-18\lambda\end{pmatrix}=\begin{pmatrix}21\\-3\\9\end{pmatrix}+\lambda\begin{pmatrix}6\\15\\-18\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}21\\-3\\9\end{pmatrix}+\lambda\begin{pmatrix}6\\15\\-18\end{pmatrix}\text{ or }l_2:\mathbf{r}=\begin{pmatrix}21\\-3\\9\end{pmatrix}+\lambda\begin{pmatrix}2\\5\\-6\end{pmatrix}$$

**26) c)**

$$\begin{pmatrix}-1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}7+2\lambda\\-1+5\lambda\\3-6\lambda\end{pmatrix}=\begin{pmatrix}-1(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)+1(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}-7-2\lambda\\-1+5\lambda\\3-6\lambda\end{pmatrix}=\begin{pmatrix}-7\\-1\\3\end{pmatrix}+\lambda\begin{pmatrix}-2\\5\\-6\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}-7\\-1\\3\end{pmatrix}+\lambda\begin{pmatrix}-2\\5\\-6\end{pmatrix}$$

**26) d)**

$$\begin{pmatrix}1&0&0\\0&-1&0\\0&0&1\end{pmatrix}\begin{pmatrix}7+2\lambda\\-1+5\lambda\\3-6\lambda\end{pmatrix}=\begin{pmatrix}1(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)-1(-1+5\lambda)+0(3-6\lambda)\\0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}7+2\lambda\\1-5\lambda\\3-6\lambda\end{pmatrix}=\begin{pmatrix}7\\1\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\-5\\-6\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}7\\1\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\-5\\-6\end{pmatrix}$$

**26) e)**

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda) \\ 0(7+2\lambda)+1(-1+5\lambda)+0(3-6\lambda) \\ -(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 3-6\lambda \\ -1+5\lambda \\ -7-2\lambda \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -7 \end{pmatrix} + \lambda\begin{pmatrix} -6 \\ 5 \\ -2 \end{pmatrix}$$

$$\therefore l_2: \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ -7 \end{pmatrix} + \lambda\begin{pmatrix} -6 \\ 5 \\ -2 \end{pmatrix}$$

**26) f)**

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 0(7+2\lambda)-(-1+5\lambda)+0(3-6\lambda) \\ 1(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 1-5\lambda \\ 7+2\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 3 \end{pmatrix} + \lambda\begin{pmatrix} -5 \\ 2 \\ -6 \end{pmatrix}$$

$$\therefore l_2: \mathbf{r} = \begin{pmatrix} 1 \\ 7 \\ 3 \end{pmatrix} + \lambda\begin{pmatrix} -5 \\ 2 \\ -6 \end{pmatrix}$$

**27) a)**

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}\begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ 7 \\ 13 \end{pmatrix}$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 9 \\ 7 \\ 13 \end{pmatrix}$

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5\times9+2\times7+1\times13}{\sqrt{5^2+2^2+1^2}\sqrt{9^2+7^2+13^2}} = \frac{72}{\sqrt{30}\sqrt{299}} = \frac{12\sqrt{8970}}{1495}$$

$$\theta = \arccos\left(\frac{12\sqrt{8970}}{1495}\right) = 40.5° \text{ (1 DP)}$$

**27) b)**

$$\begin{pmatrix} -2 & 3 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}\begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 12 \\ 20 \end{pmatrix}$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} -3 \\ 12 \\ 20 \end{pmatrix}$

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5\times(-3)+2\times12+1\times20}{\sqrt{5^2+2^2+1^2}\sqrt{(-3)^2+12^2+20^2}} = \frac{29}{\sqrt{30}\sqrt{553}} = \frac{29\sqrt{16590}}{16590}$$

$$\theta = \arccos\left(\frac{29\sqrt{16590}}{16590}\right) = 77.0° \text{ (1 DP)}$$

**27) c)**

$$\begin{pmatrix} 5 & -3 & -4 \\ 2 & 8 & -3 \\ -5 & -4 & -1 \end{pmatrix}\begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 15 \\ 23 \\ -34 \end{pmatrix}$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 15 \\ 23 \\ -34 \end{pmatrix}$

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5\times15+2\times23+1\times(-34)}{\sqrt{5^2+2^2+1^2}\sqrt{15^2+23^2+(-34)^2}} = \frac{87}{\sqrt{30}\sqrt{1910}} = \frac{29\sqrt{573}}{1910}$$

$$\theta = \arccos\left(\frac{29\sqrt{573}}{1910}\right) = 68.7° \text{ (1 DP)}$$

**28) a)**

$(2 + 4\lambda, -3 + 3\lambda, 1 + \lambda)$

**28) b)**

$(5, -9 - 3\mu, 16\mu)$

**29) a)**

$\big(-4 + (1) \times 3, 2 + (1) \times 8, 5 + (1) \times (-4)\big)$

$= (-1, 10, 1)$

**29) b)**

$\big(-4 + (-1) \times 3, 2 + (-1) \times 8, 5 + (-1) \times (-4)\big)$

$= (-7, -6, 9)$

**29) c)**

$\left(-4+(-4)\times 3, 2+(-4)\times 8, 5+(-4)\times(-4)\right)$

$=(-16,-30,21)$

**30)**

$\overrightarrow{OA}=\begin{pmatrix}2\\1\\-5\end{pmatrix}$ and $\overrightarrow{OB}=\begin{pmatrix}2+4\lambda\\2-\lambda\\-7+\lambda\end{pmatrix}$

$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}2+4\lambda\\2-\lambda\\-7+\lambda\end{pmatrix}-\begin{pmatrix}2\\1\\-5\end{pmatrix}=\begin{pmatrix}4\lambda\\1-\lambda\\-2+\lambda\end{pmatrix}$

**31)**

$\overrightarrow{OP}=\begin{pmatrix}3\\-7\\2\end{pmatrix}$ and $\overrightarrow{OQ}=\begin{pmatrix}4-3\lambda\\-1+2\lambda\\2+5\lambda\end{pmatrix}$

$\overrightarrow{QP}=\overrightarrow{OP}-\overrightarrow{OQ}=\begin{pmatrix}3\\-7\\2\end{pmatrix}-\begin{pmatrix}4-3\lambda\\-1+2\lambda\\2+5\lambda\end{pmatrix}=\begin{pmatrix}3\lambda-1\\-6-2\lambda\\-5\lambda\end{pmatrix}$

**32) a)**

$\begin{pmatrix}1\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2-\lambda\\1+5\lambda\\2\lambda-1\end{pmatrix}=0$

$\Rightarrow 1\times(2-\lambda)+1\times(1+5\lambda)+1\times(2\lambda-1)=0$

$\Rightarrow 2-\lambda+1+5\lambda+2\lambda-1=0$

$\Rightarrow 6\lambda+2=0$

$\Rightarrow \lambda=-\frac{1}{3}$

**32) b)**

$\begin{pmatrix}2\\0\\3\end{pmatrix}\cdot\begin{pmatrix}2-\lambda\\1+5\lambda\\2\lambda-1\end{pmatrix}=0$

$\Rightarrow 2\times(2-\lambda)+0\times(1+5\lambda)+3\times(2\lambda-1)=0$

$\Rightarrow 4-2\lambda+6\lambda-3=0$

$\Rightarrow 4\lambda+1=0$

$\Rightarrow \lambda=-\frac{1}{4}$

**32) c)**

$$\begin{pmatrix}-1\\3\\-5\end{pmatrix}\cdot\begin{pmatrix}2-\lambda\\1+5\lambda\\2\lambda-1\end{pmatrix}=0$$

$$\Rightarrow(-1)\times(2-\lambda)+3\times(1+5\lambda)+(-5)\times(2\lambda-1)=0$$

$$\Rightarrow -2+\lambda+3+15\lambda-10\lambda+5=0$$

$$\Rightarrow 6\lambda+6=0$$

$$\Rightarrow \lambda=-1$$

**33) a)**

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}2\\5\\-3\end{pmatrix}-\begin{pmatrix}3\\7\\-1\end{pmatrix}=\begin{pmatrix}-1\\-2\\-2\end{pmatrix}$$

$$\therefore l_1:\mathbf{r}=\begin{pmatrix}3\\7\\-1\end{pmatrix}+\lambda\begin{pmatrix}-1\\-2\\-2\end{pmatrix}$$

**33) b)**

$38=2+6\mu\Rightarrow\mu=6$

$-17=-5-2\mu\Rightarrow\mu=6$

$-15=3-3\mu\Rightarrow\mu=6$

Consistent values of $\mu$

$\therefore P(38,-17,-15)$ lies on $l_2$

**33) c)**

As $C$ lies on $l_1$, let $\overrightarrow{OC} = \begin{pmatrix} 3-\lambda \\ 7-2\lambda \\ -1-2\lambda \end{pmatrix}$

$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix} 38 \\ -17 \\ -15 \end{pmatrix} - \begin{pmatrix} 3-\lambda \\ 7-2\lambda \\ -1-2\lambda \end{pmatrix} = \begin{pmatrix} \lambda+35 \\ 2\lambda-24 \\ 2\lambda-14 \end{pmatrix}$

If $\overrightarrow{CP}$ is perpendicular to $l_2$ then $\begin{pmatrix} \lambda+35 \\ 2\lambda-24 \\ 2\lambda-14 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -2 \\ -3 \end{pmatrix} = 0$

$\Rightarrow 6(\lambda+35) - 2(2\lambda-24) - 3(2\lambda-14) = 0$

$\Rightarrow 6\lambda + 210 - 4\lambda + 48 - 6\lambda + 42 = 0$

$\Rightarrow -4\lambda + 300 = 0$

$\Rightarrow \lambda = 75$

$\therefore \overrightarrow{OC} = \begin{pmatrix} 3-(75) \\ 7-2(75) \\ -1-2(75) \end{pmatrix} = \begin{pmatrix} -72 \\ -143 \\ -151 \end{pmatrix}$

So $C(-72, -143, -151)$

**34) a)**

$\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

$\overrightarrow{OC} = 99\overrightarrow{OA} = 99\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 198 \\ -99 \\ 198 \end{pmatrix}$

$\therefore C(198, -99, 198)$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 8 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix}$

$\therefore l: \mathbf{r} = \begin{pmatrix} 198 \\ -99 \\ 198 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix}$

**34) b)**

$$\overrightarrow{OD} = \begin{pmatrix} 198+\lambda \\ -99+9\lambda \\ 198-4\lambda \end{pmatrix}$$

$$\overrightarrow{BD} = \overrightarrow{OD} - \overrightarrow{OB} = \begin{pmatrix} 198+\lambda \\ -99+9\lambda \\ 198-4\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 8 \\ -2 \end{pmatrix} = \begin{pmatrix} 195+\lambda \\ -107+9\lambda \\ 200-4\lambda \end{pmatrix}$$

If angle $ABD = 90°$, then $\overrightarrow{AB} \cdot \overrightarrow{BD} = 0$

$$\Rightarrow \begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 195+\lambda \\ -107+9\lambda \\ 200-4\lambda \end{pmatrix} = 0$$

$$\Rightarrow 1(195+\lambda) + 9(-107+9\lambda) - 4(200-4\lambda) = 0$$

$$\Rightarrow 195 + \lambda - 963 + 81\lambda - 800 + 16\lambda = 0$$

$$\Rightarrow 98\lambda - 1568 = 0$$

$$\Rightarrow \lambda = 16$$

$$\therefore \overrightarrow{OD} = \begin{pmatrix} 198+(16) \\ -99+9(16) \\ 198-4(16) \end{pmatrix} = \begin{pmatrix} 214 \\ 45 \\ 134 \end{pmatrix}$$

So $D(214, 45, 134)$

**35)**

$l_1$ and $l_2$ intersect:

$10.25 - 3\lambda = a + 2\mu$ (1)

$-6 + 4\lambda = 1 - \mu$ (2)

$5 = 7 + b\mu$ (3)

$l_1$ and $l_2$ form an angle of 120°:

$\cos 120° = \frac{(-3)\times 2 + 4\times(-1) + 0\times b}{\sqrt{(-3)^2+4^2+0^2}\sqrt{2^2+(-1)^2+b^2}}$

$\Rightarrow -\frac{1}{2} = \frac{-10}{\sqrt{25}\sqrt{5+b^2}}$

$\Rightarrow \sqrt{5+b^2} = 4$

$\Rightarrow 5 + b^2 = 16$

$\Rightarrow b^2 = 11$

$\Rightarrow b = \sqrt{11}$ as $b$ is a positive constant.

Solve (3):

$5 = 7 + \sqrt{11}\mu \Rightarrow \mu = -\frac{2}{\sqrt{11}} = -\frac{2\sqrt{11}}{11}$

Substitute into (2):

$-6 + 4\lambda = 1 - \left(-\frac{2\sqrt{11}}{11}\right)$

$\Rightarrow \lambda = \frac{77+2\sqrt{11}}{44}$

Substitute into (1):

$10.25 - 3\left(\frac{77+2\sqrt{11}}{44}\right) = a + 2\left(-\frac{2\sqrt{11}}{11}\right)$

$\Rightarrow a = \frac{110+5\sqrt{11}}{22} = 5 + \frac{5\sqrt{11}}{22}$

**36) a)**

$\overrightarrow{OA} = \begin{pmatrix}2\\1\\1\end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix}3\\-1\\-4\end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}3\\-1\\-4\end{pmatrix} - \begin{pmatrix}2\\1\\1\end{pmatrix} = \begin{pmatrix}1\\-2\\-5\end{pmatrix}$

$l: \mathbf{r} = \begin{pmatrix}2\\1\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\-2\\-5\end{pmatrix}$

**36) b)**

$\overrightarrow{OC} = \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a-2 \\ 0 \\ -2 \end{pmatrix}$

$AC$ is perpendicular to $l$ so:

$\begin{pmatrix} a-2 \\ 0 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix} = 0$

$\Rightarrow (a-2) \times 1 + 0 \times (-2) + (-2) \times (-5) = 0$

$\Rightarrow a - 2 + 0 + 10 = 0$

$\Rightarrow a = -8$

**36) c)**

$\overrightarrow{AC} = \begin{pmatrix} -8-2 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \\ -2 \end{pmatrix}$

$\overrightarrow{AB} = \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix}$

$\left|\overrightarrow{AC}\right| = \sqrt{(-10)^2 + 0^2 + (-2)^2} = 2\sqrt{26}$

$\left|\overrightarrow{AB}\right| = \sqrt{1^2 + (-2)^2 + (-5)^2} = \sqrt{30}$

Area of triangle $= \frac{1}{2} \times \left|\overrightarrow{AC}\right| \times \left|\overrightarrow{AB}\right|$

$= \frac{1}{2} \times 2\sqrt{26} \times \sqrt{30} = 2\sqrt{195}$ units$^2$

**37) a)**

$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -5 \\ -6 \end{pmatrix} - 2\begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix} = \begin{pmatrix} -1 \\ -21 \\ 12 \end{pmatrix}$

So $A(-1, -21, 12)$

**37) b)**

Let $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ be a direction vector that is perpendicular to $\begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix}$

Then $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix} = 0$

$\Rightarrow 2a + 8b - 9c = 0$

Let $a = 0$, $b = 9$, $c = 8$

Then $2 \times 0 + 8 \times 9 - 9 \times 8 = 0$ as required.

So the vector equation of a line that is perpendicular to $l$ and passes through $A$ could be $\mathbf{r} = \begin{pmatrix} -1 \\ -21 \\ 12 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 9 \\ 8 \end{pmatrix}$

**38) a)**

Note that $\begin{pmatrix}-4\\1\\5\end{pmatrix} = -\begin{pmatrix}4\\-1\\-5\end{pmatrix}$ so $l_1$ and $l_3$ are parallel

and that $\begin{pmatrix}-2\\0\\1\end{pmatrix} = \frac{1}{3}\begin{pmatrix}-6\\0\\3\end{pmatrix}$ so $l_2$ and $l_4$ are parallel

So the quadrilateral has two pairs of parallel sides, and so is either a parallelogram or a rhombus.

Find intersection of $l_1$ & $l_2$:

$-7-4\alpha = 3-2\beta \Rightarrow -4\alpha+2\beta = 10$ (1)

$3+\alpha = 2 \Rightarrow \alpha = -1$ (2)

$10+5\alpha = 2+\beta \Rightarrow 5\alpha-\beta = -8$ (3)

Solve (1) and (2):

$\alpha = -1$ and $\beta = 3$

Check in (3):

$5(-1)-(3) = -8$ as required.

Substitute $\alpha = -1$ into $l_1$: $\mathbf{r} = \begin{pmatrix}-7\\3\\10\end{pmatrix} + (-1)\begin{pmatrix}-4\\1\\5\end{pmatrix} = \begin{pmatrix}-3\\2\\5\end{pmatrix}$

So the 1st vertex has coordinates $(-3, 2, 5)$

Find intersection of $l_1$ & $l_4$:

$-7-4\alpha = 19-6\delta \Rightarrow -4\alpha+6\delta = 26$ (1)

$3+\alpha = 1 \Rightarrow \alpha = -2$ (2)

$10+5\alpha = -9+3\delta \Rightarrow 5\alpha-3\delta = -19$ (3)

Solve (1) and (2):

$\alpha = -2$ and $\delta = 3$

Check in (3):

$5(-2)-3(3) = -19$ as required.

Substitute $\alpha = -2$ into $l_1$: $\mathbf{r} = \begin{pmatrix}-7\\3\\10\end{pmatrix} + (-2)\begin{pmatrix}-4\\1\\5\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix}$

So the 2nd vertex has coordinates $(1, 1, 0)$

Find intersection of $l_2$ & $l_3$:

$3-2\beta = -21+4\gamma \Rightarrow -2\beta-4\gamma = -24$ (1)

$2 = 5-\gamma \Rightarrow \gamma = 3$ (2)

$2 + \beta = 23 - 5\gamma \Rightarrow \beta + 5\gamma = 21$ (3)

Solve (1) and (2):

$\beta = 6$ and $\gamma = 3$

Check in (3):

$(6) + 5(3) = 21$ as required.

Substitute $\beta = 6$ into $l_2$: $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} + (6)\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -9 \\ 2 \\ 8 \end{pmatrix}$

So the 3rd vertex has coordinates $(-9, 2, 8)$

Now let $A(-3, 2, 5)$, $B(1, 1, 0)$ and $C(-9, 2, 8)$ be three of the four vertices of the parallelogram.

$AB = \sqrt{(1+3)^2 + (1-2)^2 + (0-5)^2} = \sqrt{42}$

$AC = \sqrt{(-9+3)^2 + (2-2)^2 + (8-5)^2} = \sqrt{45}$

$AB \neq AC$ so it is not a rhombus

**38) b)**

$\cos\theta = \frac{\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}}{\left|\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}\right| \left|\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}\right|} = \frac{(-4)\times(-2)+1\times 0+5\times 1}{\sqrt{(-4)^2+1^2+5^2}\sqrt{(-2)^2+0^2+1^2}} = \frac{13}{\sqrt{42}\sqrt{5}} = \frac{13}{\sqrt{210}}$

$\sin\theta = \frac{\sqrt{\left(\sqrt{210}\right)^2 - 13^2}}{\sqrt{210}} = \frac{\sqrt{41}}{\sqrt{210}}$

Area of triangle $= \frac{1}{2} \times \sqrt{42} \times \sqrt{45} \times \frac{\sqrt{41}}{\sqrt{210}} = \frac{3\sqrt{41}}{2}$ units$^2$

So area of parallelogram is $3\sqrt{41}$ units$^2$

**39) a)**

$\overrightarrow{OA} = \begin{pmatrix} 6 \\ -6 \\ 1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 14 \\ -12 \\ 5 \end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 14 \\ -12 \\ 5 \end{pmatrix} - \begin{pmatrix} 6 \\ -6 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -6 \\ 4 \end{pmatrix}$

$l_1: r = \begin{pmatrix} 6 \\ -6 \\ 1 \end{pmatrix} + \lambda\begin{pmatrix} 8 \\ -6 \\ 4 \end{pmatrix}$

**39) b)**

$6 + 8\lambda = 0 + 3\mu \Rightarrow 8\lambda - 3\mu = -6$ (1)

$-6 - 6\lambda = -2 - 2\mu \Rightarrow -6\lambda + 2\mu = 4$ (2)

$1 + 4\lambda = 3 - \mu \Rightarrow 4\lambda + \mu = 2$ (3)

Solve (1) and (2): $\lambda = 0$ and $\mu = 2$

Check in (3):

$4(0) + (2) = 2$ as required

So the lines intersect.

**39) c)**

$D\big(0 + (4) \times 3, -2 + (4) \times (-2), 3 + (4) \times (-1)\big)$

$= D(12, -10, -1)$

**39) d)**

The line through $D$ and $C$ is parallel to $l_1$ and so has the equation

$$\mathbf{r} = \begin{pmatrix} 12 \\ -10 \\ -1 \end{pmatrix} + \alpha \begin{pmatrix} 8 \\ -6 \\ 4 \end{pmatrix}$$

So $C$ has coordinates $(12 + 8\alpha, -10 - 6\alpha, -1 + 4\alpha)$ for some $\alpha$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 12 \\ -10 \\ -1 \end{pmatrix} - \begin{pmatrix} 6 \\ -6 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ -2 \end{pmatrix}$$

$$\left|\overrightarrow{AD}\right| = \sqrt{6^2 + (-4)^2 + (-2)^2} = \sqrt{56} = 2\sqrt{14}$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} 12 + 8\alpha \\ -10 - 6\alpha \\ -1 + 4\alpha \end{pmatrix} - \begin{pmatrix} 14 \\ -12 \\ 5 \end{pmatrix} = \begin{pmatrix} 8\alpha - 2 \\ 2 - 6\alpha \\ 4\alpha - 6 \end{pmatrix}$$

$$\left|\overrightarrow{BC}\right| = \left|\overrightarrow{AD}\right| = \sqrt{56}$$

$$\therefore \sqrt{(8\alpha - 2)^2 + (2 - 6\alpha)^2 + (4\alpha - 6)^2} = \sqrt{56}$$

$$\Rightarrow 64\alpha^2 - 32\alpha + 4 + 4 - 24\alpha + 36\alpha^2 + 16\alpha^2 - 48\alpha + 36 = 56$$

$$\Rightarrow 64\alpha^2 - 32\alpha + 4 + 4 - 24\alpha + 36\alpha^2 + 16\alpha^2 - 48\alpha + 36 = 56$$

$$\Rightarrow 116\alpha^2 - 104\alpha - 12 = 0$$

$$\Rightarrow 29\alpha^2 - 26\alpha - 3 = 0$$

$$\Rightarrow (29\alpha + 3)(\alpha - 1) = 0$$

$$\Rightarrow \alpha = -\frac{3}{29} \text{ or } \alpha = 1$$

So either $C\left(12 + 8\left(-\frac{3}{29}\right), -10 - 6\left(-\frac{3}{29}\right), -1 + 4\left(-\frac{3}{29}\right)\right) = C\left(\frac{324}{29}, -\frac{272}{29}, -\frac{41}{29}\right)$

Or $C\big(12 + 8(1), -10 - 6(1), -1 + 4(1)\big) = C(20, -16, 3)$

If $\alpha = -\frac{3}{29}$ then $\overrightarrow{BC} = \begin{pmatrix} 8\left(-\frac{3}{29}\right) - 2 \\ 2 - 6\left(-\frac{3}{29}\right) \\ 4\left(-\frac{3}{29}\right) - 6 \end{pmatrix} = \begin{pmatrix} -\frac{82}{29} \\ \frac{76}{29} \\ -\frac{186}{29} \end{pmatrix} = -\frac{2}{29}\begin{pmatrix} 41 \\ 38 \\ 93 \end{pmatrix}$

If $\alpha = 1$ then $\overrightarrow{BC} = \begin{pmatrix} 8(1) - 2 \\ 2 - 6(1) \\ 4(1) - 6 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ -2 \end{pmatrix} = \overrightarrow{AD}$

So if $\alpha = 1$, then $ABCD$ is a parallelogram and not a trapezium.

So $\alpha = -\frac{3}{29}$ and hence $C\left(\frac{324}{29}, -\frac{272}{29}, -\frac{41}{29}\right)$

**40) a)**

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix} - \begin{pmatrix} 4 \\ 7 \\ -6 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}$

$\therefore l_1: \mathbf{r} = \begin{pmatrix} 4 \\ 7 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}$

$\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \begin{pmatrix} 3 \\ 3 \\ -9 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}$

$\therefore l_2: \mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}$

$4 - 3\lambda = 3$ (1)

$7 - 3\lambda = 5 - 2\mu \Rightarrow -3\lambda + 2\mu = -2$ (2)

$-6 = -7 - 2\mu$ (3)

Solve (1) and (3): $\lambda = \frac{1}{3}$ and $\mu = -\frac{1}{2}$

Check in (2):

$-3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) = -2$ as required

So the lines intersect.

Substitute $\lambda = \frac{1}{3}$ into $l_1$:

So $P$ has coordinates $(3, 6, -6)$

**40) b)**

$\overrightarrow{BP} = \overrightarrow{OP} - \overrightarrow{OB} = \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$

$\therefore \left|\overrightarrow{BP}\right| = \sqrt{2^2 + 2^2 + 0^2} = \sqrt{8} = 2\sqrt{2}$

$E$ lies on $l_2$ so it has coordinates $(3, 5 - 2\mu, -7 - 2\mu)$ for some value $\mu$

$\overrightarrow{PE} = \overrightarrow{OE} - \overrightarrow{OP} = \begin{pmatrix} 3 \\ 5 - 2\mu \\ -7 - 2\mu \end{pmatrix} - \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 - 2\mu \\ -1 - 2\mu \end{pmatrix}$

$\left|\overrightarrow{PE}\right| = \sqrt{8}$

$\Rightarrow \sqrt{0^2 + (-1 - 2\mu)^2 + (-1 - 2\mu)^2} = \sqrt{8}$

$\Rightarrow 0 + 1 + 4\mu + 4\mu^2 + 1 + 4\mu + 4\mu^2 = 8$

$\Rightarrow 8\mu^2 + 8\mu - 6 = 0$

$\Rightarrow 4\mu^2 + 4\mu - 3 = 0$

$\Rightarrow (2\mu + 3)(2\mu - 1) = 0$

$\Rightarrow \mu = -\frac{3}{2}$ or $\mu = \frac{1}{2}$

If $E$ has coordinates $\left(3, 5 - 2\left(-\frac{3}{2}\right), -7 - 2\left(-\frac{3}{2}\right)\right) = (3, 8, -4)$

Or $E$ has coordinates $\left(3, 5 - 2\left(\frac{1}{2}\right), -7 - 2\left(\frac{1}{2}\right)\right) = (3, 4, -8)$

**41)**

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4+\sqrt{6} \\ -5-3\sqrt{6} \\ -2-\sqrt{6} \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} = \begin{pmatrix} \sqrt{6} \\ -3\sqrt{6} \\ -6 \end{pmatrix}$$

$$\therefore l_1: \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} + \lambda \begin{pmatrix} \sqrt{6} \\ -3\sqrt{6} \\ -6 \end{pmatrix}$$

$$\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \begin{pmatrix} \sqrt{6}-9 \\ 1-6\sqrt{6} \\ 5-2\sqrt{6} \end{pmatrix} - \begin{pmatrix} 3+\sqrt{6} \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -12 \\ -6\sqrt{6} \\ -2\sqrt{6} \end{pmatrix}$$

$$\therefore l_2: \mathbf{r} = \begin{pmatrix} 3+\sqrt{6} \\ 1 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -12 \\ -6\sqrt{6} \\ -2\sqrt{6} \end{pmatrix}$$

$4 + \sqrt{6}\lambda = 3 + \sqrt{6} - 12\mu \Rightarrow \sqrt{6}\lambda + 12\mu = -1 + \sqrt{6}$ (1)

$-5 - 3\sqrt{6}\lambda = 1 - 6\sqrt{6}\mu \Rightarrow -3\sqrt{6}\lambda + 6\sqrt{6}\mu = 6$ (2)

$4 - \sqrt{6} - 6\lambda = 5 - 2\sqrt{6}\mu \Rightarrow -6\lambda + 2\sqrt{6}\mu = 1 + \sqrt{6}$ (3)

Solve (1) and (2): $\lambda = -\frac{\sqrt{6}}{6}$ and $\mu = \frac{\sqrt{6}}{12}$

Check in (3):

$-6\left(-\frac{\sqrt{6}}{6}\right) + 2\sqrt{6}\left(\frac{\sqrt{6}}{12}\right) = 1 + \sqrt{6}$ as required

So the lines intersect.

Substitute $\lambda = -\frac{\sqrt{6}}{6}$ into $l_1$:

So the centre of the sphere, $P$, has coordinates $(3, -2, 4)$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ \sqrt{6} \end{pmatrix}$$

$$\left|\overrightarrow{AP}\right| = \sqrt{(-1)^2 + 3^2 + \left(\sqrt{6}\right)^2} = \sqrt{16} = 4$$

Volume $= \frac{4}{3}\pi \times 4^3 = \frac{256\pi}{3}$ units$^3$

**42) a)**

$3 + 3\lambda = 5 + 16\mu \Rightarrow 3\lambda - 16\mu = 2$ (1)

$2 - 3\lambda = -8 - 8\mu \Rightarrow -3\lambda + 8\mu = -10$ (2)

$-4 + 5\lambda = 3 + 23\mu \Rightarrow 5\lambda - 23\mu = 7$ (3)

Solve (1) and (2): $\lambda = 6$ and $\mu = 1$

Check in (3):

$5(6) - 23(1) = 7$ as required

So the lines intersect.

Substitute $\lambda = 6$ into $l_1$:

The lines intersect at $P(21, -16, 26)$

**42) b)**

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 21 \\ -16 \\ 26 \end{pmatrix} - \begin{pmatrix} 3+3\lambda \\ 2-3\lambda \\ -4+5\lambda \end{pmatrix} = \begin{pmatrix} 18-3\lambda \\ -18+3\lambda \\ 30-5\lambda \end{pmatrix}$

$\left|\overrightarrow{AP}\right| = 849\sqrt{43} \Rightarrow \sqrt{(18-3\lambda)^2 + (-18+3\lambda)^2 + (30-5\lambda)^2} = 849\sqrt{43}$

$\Rightarrow 324 - 108\lambda + 9\lambda^2 + 324 - 108\lambda + 9\lambda^2 + 900 - 300\lambda + 25\lambda^2 = 30994443$

$\Rightarrow 43\lambda^2 - 516\lambda - 30992895 = 0$

$\Rightarrow \lambda^2 - 12\lambda - 720765 = 0$

$\Rightarrow (\lambda + 843)(\lambda - 855) = 0$

$\Rightarrow \lambda = -843$ or $\lambda = 855$

So $A$ has coordinates $\big(3+3(-843), 2-3(-843), -4+5(-843)\big)$

$= (-2526, 2531, -4219)$

or $\big(3+3(855), 2-3(855), -4+5(855)\big) = (2568, -2563, 4271)$

If $A(-2526, 2531, -4219)$ then:

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5+16\mu \\ -8-8\mu \\ 3+23\mu \end{pmatrix} - \begin{pmatrix} -2526 \\ 2531 \\ -4219 \end{pmatrix} = \begin{pmatrix} 2531+16\mu \\ -2539-8\mu \\ 4222+23\mu \end{pmatrix}$

$\begin{pmatrix} 16 \\ -8 \\ 23 \end{pmatrix} \cdot \begin{pmatrix} 2531+16\mu \\ -2539-8\mu \\ 4222+23\mu \end{pmatrix} = 0$

$\Rightarrow 16 \times (2531+16\mu) + (-8) \times (-2539-8\mu) + 23 \times (4222+23\mu) = 0$

$\Rightarrow 40496 + 256\mu + 20312 + 64\mu + 97106 + 529\mu = 0$

$\Rightarrow 849\mu + 157914 = 0$

$\Rightarrow \mu = -186$

So $B$ has coordinates $\big(5+16(-186), -8-8(-186), 3+23(-186)\big) =$

$(-2971, 1480, -4275)$

If $A(2568, -2563, 4271)$ then:

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5+16\mu \\ -8-8\mu \\ 3+23\mu \end{pmatrix} - \begin{pmatrix} 2568 \\ -2563 \\ 4271 \end{pmatrix} = \begin{pmatrix} -2563+16\mu \\ 2555-8\mu \\ -4268+23\mu \end{pmatrix}$

$\begin{pmatrix} 16 \\ -8 \\ 23 \end{pmatrix} \cdot \begin{pmatrix} -2563+16\mu \\ 2555-8\mu \\ -4268+23\mu \end{pmatrix} = 0$

$\Rightarrow 16 \times (-2563+16\mu) + (-8) \times (2555-8\mu) + 23 \times (-4268+23\mu) = 0$

$\Rightarrow -41008 + 256\mu - 20440 + 64\mu - 98164 + 529\mu = 0$

$\Rightarrow 849\mu - 159612 = 0$

$\Rightarrow \mu = 188$

So $B$ has coordinates $(5+16(188), -8-8(188), 3+23(188))$
$= (3013, -1512, 4327)$
So either $B(-2971, 1480, -4275)$ or $B(3013, -1512, 4327)$

**43) a)**

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}-1\\3\\-2\end{pmatrix} - \begin{pmatrix}3\\4\\-5\end{pmatrix} = \begin{pmatrix}-4\\-1\\3\end{pmatrix}$$

$$\therefore l: \mathbf{r} = \begin{pmatrix}3\\4\\-5\end{pmatrix} + \lambda\begin{pmatrix}-4\\-1\\3\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}3\\p\\q\end{pmatrix} - \begin{pmatrix}3\\4\\-5\end{pmatrix} = \begin{pmatrix}0\\p-4\\q+5\end{pmatrix}$$

$\left|\overrightarrow{AC}\right| = 90 \Rightarrow \sqrt{0^2 + (p-4)^2 + (q+5)^2} = \sqrt{90}$
$\Rightarrow (p-4)^2 + (q+5)^2 = 90$ (1)

$$\begin{pmatrix}-4\\-1\\3\end{pmatrix} \cdot \begin{pmatrix}0\\p-4\\q+5\end{pmatrix} = 0$$

$\Rightarrow (-4) \times 0 + (-1) \times (p-4) + 3 \times (q+5) = 0$
$\Rightarrow 0 - p + 4 + 3q + 15 = 0$
$\Rightarrow p = 3q + 19$ (2)
Substitute (2) into (1):
$(3q+15)^2 + (q+5)^2 = 90$
$\Rightarrow 9q^2 + 90q + 225 + q^2 + 10q + 25 = 90$
$\Rightarrow 10q^2 + 100q + 160 = 0$
$\Rightarrow q^2 + 10q + 16 = 0$
$\Rightarrow (q+8)(q+2) = 0$
$\Rightarrow q = -8$ or $q = -2$
If $q = -8$ then $p = 3(-8) + 19 = -5$
If $q = -2$ then $p = 3(-2) + 19 = 13$
As $p > 0$, we must have that $C(3, 13, -2)$

**43) b)**

$\left|\overrightarrow{AB}\right| = \sqrt{(-4)^2 + (-1)^2 + 3^2} = \sqrt{26}$

$l: \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$

$D$ has coordinates $(3 - 4\lambda, 4 - \lambda, -5 + 3\lambda)$ for some $\lambda$

$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 3 - 4\lambda \\ 4 - \lambda \\ -5 + 3\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -4\lambda \\ -\lambda \\ 3\lambda \end{pmatrix}$

$\left|\overrightarrow{AD}\right| = \sqrt{(-4\lambda)^2 + (-\lambda)^2 + (3\lambda)^2} = \sqrt{26\lambda^2}$

Area of triangle $CAB = \frac{1}{2} \times \left|\overrightarrow{AC}\right| \times \left|\overrightarrow{AB}\right| = \frac{1}{2} \times \sqrt{90} \times \sqrt{26}$

Area of triangle $CAD = \frac{1}{2} \times \left|\overrightarrow{AC}\right| \times \left|\overrightarrow{AD}\right| = \frac{1}{2} \times \sqrt{90} \times \sqrt{26\lambda^2}$

$\frac{1}{2} \times \sqrt{90} \times \sqrt{26\lambda^2} = 2 \times \frac{1}{2} \times \sqrt{90} \times \sqrt{26}$

$\Rightarrow \sqrt{26\lambda^2} = 2\sqrt{26}$

$\Rightarrow 26\lambda^2 = 4 \times 26$

$\Rightarrow \lambda^2 = 4$

$\Rightarrow \lambda = \pm 2$

So $D$ has coordinates $\big(3 - 4(-2), 4 - (-2), -5 + 3(-2)\big) = (11, 6, -11)$

Or $D$ has coordinates $\big(3 - 4(2), 4 - (2), -5 + 3(2)\big) = (-5, 2, 1)$

**Alternative Method:**

$$\overrightarrow{OD} = \overrightarrow{OA} \pm 2\overrightarrow{AB}$$

$= \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} \pm 2 \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 11 \\ 6 \\ -11 \end{pmatrix}$

So $D$ has coordinates $(-5, 2, 1)$ or $(11, 6, -11)$

**44) a)**

$\left(\frac{-1+3\sqrt{89}-1-3\sqrt{89}}{2}, \frac{5-2\sqrt{89}+5+2\sqrt{89}}{2}, \frac{9+6\sqrt{89}+9-6\sqrt{89}}{2}\right) = (-1, 5, 9)$

**44) b)**

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 - 3\sqrt{89} \\ 5 + 2\sqrt{89} \\ 9 - 6\sqrt{89} \end{pmatrix} - \begin{pmatrix} -1 + 3\sqrt{89} \\ 5 - 2\sqrt{89} \\ 9 + 6\sqrt{89} \end{pmatrix} = \begin{pmatrix} -6\sqrt{89} \\ 4\sqrt{89} \\ -12\sqrt{89} \end{pmatrix}$$

$$l_1 : r = \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} -6\sqrt{89} \\ 4\sqrt{89} \\ -12\sqrt{89} \end{pmatrix}$$

To make the numbers nicer, we'll choose to use $l_1 : r = \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix}$

$-1 + 3\lambda = 3 + 2\mu \Rightarrow 3\lambda - 2\mu = 4$ (1)

$5 - 2\lambda = 23 + 9\mu \Rightarrow -2\lambda - 9\mu = 18$ (2)

$9 + 6\lambda = 13 + 2\mu \Rightarrow 6\lambda - 2\mu = 4$ (3)

Solve (1) and (2): $\lambda = 0$ and $\mu = -2$

Check in (3):

$6(0) - 2(-2) = 4$ as required

So the lines intersect.

Substitute $\lambda = 0$ into $l_1$:

The lines intersect at $(-1, 5, 9)$

**44) c)**

$\left|\overrightarrow{AP}\right| = \frac{1}{2}\left|\overrightarrow{AB}\right| = \frac{1}{2}\sqrt{\left(-6\sqrt{89}\right)^2 + \left(4\sqrt{89}\right)^2 + \left(-12\sqrt{89}\right)^2}$

$= \frac{1}{2}\sqrt{3204 + 1424 + 12816} = \frac{1}{2}\sqrt{17444} = 7\sqrt{89}$

$\overrightarrow{PC} = \overrightarrow{OC} - \overrightarrow{OP} = \begin{pmatrix} 3+2\mu \\ 23+9\mu \\ 13+2\mu \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} = \begin{pmatrix} 4+2\mu \\ 18+9\mu \\ 4+2\mu \end{pmatrix}$

$\left|\overrightarrow{PC}\right| = 7\sqrt{89}$

$\Rightarrow \sqrt{(4+2\mu)^2 + (18+9\mu)^2 + (4+2\mu)^2} = 7\sqrt{89}$

$\Rightarrow 16 + 16\mu + 4\mu^2 + 324 + 324\mu + 81\mu^2 + 16 + 16\mu + 4\mu^2 = 4361$

$\Rightarrow 89\mu^2 + 356\mu - 4005 = 0$

$\Rightarrow \mu^2 + 4\mu - 45 = 0$

$\Rightarrow (\mu + 9)(\mu - 5) = 0$

$\Rightarrow \mu = -9$ or $\mu = 5$

So $C$ and $D$ have coordinates $\big(3 + 2(-9), 23 + 9(-9), 13 + 2(-9)\big) = (-15, -58, -5)$

or $\big(3 + 2(5), 23 + 9(5), 13 + 2(5)\big) = (13, 68, 23)$

**45)**

$P(18, 4, -1)$ lies on $l_3$, and $l_3$ is parallel to $l_1$ so:

$l_3: \mathbf{r} = \begin{pmatrix} 18 \\ 4 \\ -1 \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$

$Q$ lies on $l_3$ and so has coordinates $(18 + 4\alpha, 4 - 2\alpha, -1 + \alpha)$ for some $\alpha$

$R$ lies on the line $l_2$ when $\mu = -27$ and so has coordinates

$(0 + (-27) \times (-6), 7 + (-27) \times 1, 5 + (-27) \times 2) = (162, -20, -49)$

$\overrightarrow{RQ} = \overrightarrow{OQ} - \overrightarrow{OR} = \begin{pmatrix} 18+4\alpha \\ 4-2\alpha \\ -1+\alpha \end{pmatrix} - \begin{pmatrix} 162 \\ -20 \\ -49 \end{pmatrix} = \begin{pmatrix} -144+4\alpha \\ 24-2\alpha \\ 48+\alpha \end{pmatrix}$

$\overrightarrow{RQ}$ is perpendicular to $l_2$ so:

$\begin{pmatrix} -6 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -144+4\alpha \\ 24-2\alpha \\ 48+\alpha \end{pmatrix} = 0$

$\Rightarrow (-6) \times (-144 + 4\alpha) + 1 \times (24 - 2\alpha) + 2 \times (48 + \alpha) = 0$

$\Rightarrow 864 - 24\alpha + 24 - 2\alpha + 96 + 2\alpha = 0$

$\Rightarrow 984 - 24\alpha = 0$

$\Rightarrow \alpha = 41$

$\therefore Q$ has coordinates $\big(18 + 4(41), 4 - 2(41), -1 + (41)\big) = (182, -78, 40)$

**46) a)**

Drone 1 travels from $(0, 12, 1)$ to $(15, 6, 15)$

$$\begin{pmatrix}15\\6\\15\end{pmatrix} - \begin{pmatrix}0\\12\\1\end{pmatrix} = \begin{pmatrix}15\\-6\\14\end{pmatrix}$$

$$l_1: \mathbf{r} = \begin{pmatrix}0\\12\\1\end{pmatrix} + \lambda\begin{pmatrix}15\\-6\\14\end{pmatrix}$$

Drone 2 travels from $(0, 3, 4)$ to $(15, 9, k)$

$$\begin{pmatrix}15\\9\\k\end{pmatrix} - \begin{pmatrix}0\\3\\4\end{pmatrix} = \begin{pmatrix}15\\6\\k-4\end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix}0\\3\\4\end{pmatrix} + \mu\begin{pmatrix}15\\6\\k-4\end{pmatrix}$$

$0 + 15\lambda = 0 + 15\mu \Rightarrow \lambda = \mu$ (1)

$12 - 6\lambda = 3 + 6\mu \Rightarrow -6\lambda - 6\mu = -9$ (2)

$1 + 14\lambda = 4 + \mu(k - 4)$ (3)

Solve (1) and (2): $\lambda = \frac{3}{4}$ and $\mu = \frac{3}{4}$

Substitute into (3):

$1 + 14\left(\frac{3}{4}\right) = 4 + \left(\frac{3}{4}\right)(k - 4)$

$\Rightarrow 1 + \frac{21}{2} = 4 + \frac{3}{4}k - 3$

$\Rightarrow k = 14$

**46) b)**

Distance from $(0, 12, 1)$ to $(15, 6, 15)$ is $\sqrt{(15-0)^2 + (6-12)^2 + (15-1)^2}$

$= \sqrt{457}$ metres

So Drone 1 travels $\sqrt{457}$ metres in 5 seconds, so it is travelling at a constant speed of $\frac{1}{5}\sqrt{457}$ ms$^{-1}$

Point of intersection of $l_1$ and $l_2$:

$\left(0 + \left(\frac{3}{4}\right) \times 15, 12 + \left(\frac{3}{4}\right) \times -6, 1 + \left(\frac{3}{4}\right) \times 14\right) = \left(\frac{45}{4}, \frac{15}{2}, \frac{23}{2}\right)$

Distance from $(0, 12, 1)$ to $\left(\frac{45}{4}, \frac{15}{2}, \frac{23}{2}\right)$ is $\sqrt{\left(\frac{45}{4} - 0\right)^2 + \left(\frac{15}{2} - 12\right)^2 + \left(\frac{23}{2} - 1\right)^2}$

$= \sqrt{\frac{4113}{4}} = \frac{3}{4}\sqrt{457}$ metres

Drone 1 travels from its initial starting position to the point of intersection in $\frac{\frac{3}{4}\sqrt{457}}{\frac{1}{5}\sqrt{457}} = \frac{15}{4}$ seconds

Distance from $(0, 3, 4)$ to $\left(\frac{45}{4}, \frac{15}{2}, \frac{23}{2}\right)$ is $\sqrt{\left(\frac{45}{4} - 0\right)^2 + \left(\frac{15}{2} - 3\right)^2 + \left(\frac{23}{2} - 4\right)^2}$

$= \sqrt{\frac{3249}{16}} = \frac{57}{4}$ metres

So Drone 2 must travel $\frac{57}{4}$ metres in $\frac{15}{4}$ seconds to crash into Drone 1, so its speed must be $\frac{\frac{57}{4}}{\frac{15}{4}} = \frac{19}{5} = 3.8$ ms$^{-1}$

**46) c)**

The drones are modelled as particles, with negligible size, so they could crash into one another sooner. We have also assumed that the drones instantaneously reach their constant speed.

**47) a)**

$$\begin{pmatrix}9\\3\\-3\end{pmatrix}-\begin{pmatrix}6\\2\\-2\end{pmatrix}=\begin{pmatrix}3\\1\\-1\end{pmatrix}$$

$$\begin{pmatrix}6\\2\\-2\end{pmatrix}-2\begin{pmatrix}3\\1\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

So the origin lies on the line through $A$ and $B$

So $\mathbf{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\1\\-1\end{pmatrix}=\lambda\begin{pmatrix}3\\1\\-1\end{pmatrix}$ as required

**47) b)**

$$\overrightarrow{OC}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&1\end{pmatrix}\begin{pmatrix}6\\2\\-2\end{pmatrix}=\begin{pmatrix}-2\\6\\-2\end{pmatrix}$$

$$\overrightarrow{OD}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&1\end{pmatrix}\begin{pmatrix}9\\3\\-3\end{pmatrix}=\begin{pmatrix}-3\\9\\-3\end{pmatrix}$$

$$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=\begin{pmatrix}-3\\9\\-3\end{pmatrix}-\begin{pmatrix}-2\\6\\-2\end{pmatrix}=\begin{pmatrix}-1\\3\\-1\end{pmatrix}$$

$$\begin{pmatrix}-2\\6\\-2\end{pmatrix}-2\begin{pmatrix}-1\\3\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

So the origin lies on the line through $C$ and $D$

So $\mathbf{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\-1\end{pmatrix}=\mu\begin{pmatrix}-1\\3\\-1\end{pmatrix}$

**47) c)**

$$\cos\theta=\frac{\begin{pmatrix}3\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\3\\-1\end{pmatrix}}{\left|\begin{pmatrix}3\\1\\-1\end{pmatrix}\right|\left|\begin{pmatrix}-1\\3\\-1\end{pmatrix}\right|}=\frac{3\times(-1)+1\times3+(-1)\times(-1)}{\sqrt{3^2+1^2+(-1)^2}\sqrt{(-1)^2+3^2+(-1)^2}}$$

$$=\frac{1}{\sqrt{11}\sqrt{11}}=\frac{1}{11}$$

$\theta=\arccos\left(\frac{1}{11}\right)=84.8°$ (1 DP)

**48)**

$$\begin{pmatrix}1 & 2 & 0\\ -1 & 5 & 0\\ 0 & 0 & 3\end{pmatrix}\begin{pmatrix}1+2\lambda\\ 1-\lambda\\ 2+3\lambda\end{pmatrix}=\begin{pmatrix}1(1+2\lambda)+2(1-\lambda)+0(2+3\lambda)\\ -(1+2\lambda)+5(1-\lambda)+0(2+3\lambda)\\ 0(1+2\lambda)+0(1-\lambda)+3(2+3\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}3\\ 4-7\lambda\\ 6+9\lambda\end{pmatrix}=\begin{pmatrix}3\\ 4\\ 6\end{pmatrix}+\lambda\begin{pmatrix}0\\ -7\\ 9\end{pmatrix}$$

So $l_2$: $\mathbf{r}=\begin{pmatrix}3\\ 4\\ 6\end{pmatrix}+\lambda\begin{pmatrix}0\\ -7\\ 9\end{pmatrix}$

**49)**

$$\begin{pmatrix}2 & 1 & 0\\ 1 & 3 & 1\\ 1 & 0 & 2\end{pmatrix}\begin{pmatrix}2+3\lambda\\ 3-\lambda\\ -4-\lambda\end{pmatrix}=\begin{pmatrix}2(2+3\lambda)+1(3-\lambda)+0(-4-\lambda)\\ 1(2+3\lambda)+3(3-\lambda)+1(-4-\lambda)\\ 1(2+3\lambda)+0(3-\lambda)+2(-4-\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}7+5\lambda\\ 7-\lambda\\ -6+\lambda\end{pmatrix}=\begin{pmatrix}7\\ 7\\ -6\end{pmatrix}+\lambda\begin{pmatrix}5\\ -1\\ 1\end{pmatrix}$$

So $l_2$: $\mathbf{r}=\begin{pmatrix}7\\ 7\\ -6\end{pmatrix}+\lambda\begin{pmatrix}5\\ -1\\ 1\end{pmatrix}$

**50)**

$$\begin{pmatrix}1 & -2 & 1\\ 1.5 & -1 & 0.25\\ 1 & 0.5 & -3\end{pmatrix}\begin{pmatrix}5+\lambda\\ 4-\lambda\\ 2+2\lambda\end{pmatrix}=\begin{pmatrix}1(5+\lambda)-2(4-\lambda)+1(2+2\lambda)\\ 1.5(5+\lambda)-(4-\lambda)+0.25(2+2\lambda)\\ 1(5+\lambda)+0.5(4-\lambda)-3(2+2\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}5+\lambda-8+2\lambda+2+2\lambda\\ \frac{15}{2}+\frac{3}{2}\lambda-4+\lambda+\frac{1}{2}+\frac{1}{2}\lambda\\ 5+\lambda+2-\frac{1}{2}\lambda-6-6\lambda\end{pmatrix}=\begin{pmatrix}5\lambda-1\\ 4+3\lambda\\ 1-\frac{11}{2}l\end{pmatrix}=\begin{pmatrix}-1\\ 4\\ 1\end{pmatrix}+\lambda\begin{pmatrix}5\\ 3\\ -\frac{11}{2}\end{pmatrix}$$

$$=\begin{pmatrix}-1\\ 4\\ 1\end{pmatrix}+\mu\begin{pmatrix}10\\ 6\\ -11\end{pmatrix}$$

Where $\mu=\frac{1}{2}\lambda$

**51)**

$$\begin{pmatrix} p & 1 & -3.5 \\ 3 & -3 & 2 \\ 2.5 & -1 & q \end{pmatrix}\begin{pmatrix} 2+8\lambda \\ 3+2\lambda \\ -6-5\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} p(2+8\lambda)+1(3+2\lambda)-\frac{7}{2}(-6-5\lambda) \\ 3(2+8\lambda)-3(3+2\lambda)+2(-6-5\lambda) \\ \frac{5}{2}(2+8\lambda)-(3+2\lambda)+q(-6-5\lambda) \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2p+8p\lambda+3+2\lambda+21+\frac{35}{2}\lambda \\ 6+24\lambda-9-6\lambda-12-10\lambda \\ 5+20\lambda-3-2\lambda-6q-5q\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (2p+24)+\left(8p+\frac{39}{2}\right)\lambda \\ -15+8\lambda \\ (2-6q)+(18-5q)\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$(2p+24)+\left(8p+\frac{39}{2}\right)\lambda = 26+a\mu$ (1)

$-15+8\lambda = -15+16\mu \Rightarrow \lambda = 2\mu$ (2)

$(2-6q)+(18-5q)\lambda = -10+b\mu$ (3)

Substitute (2) into (1):

$(2p+24)+(16p+39)\mu = 26+a\mu$

$2p+24 = 26 \Rightarrow p = 1$

$16\times 1+39 = a \Rightarrow a = 55$

Substitute (2) into (3)

$(2-6q)+(36-10q)\mu = -10+b\mu$

$2-6q = -10 \Rightarrow q = 2$

$36-10\times 2 = b \Rightarrow b = 16$

So $a = 55$, $b = 16$, $p = 1$, $q = 2$

**52)**

Let $C$ be the centre of the circle, so $\overrightarrow{OC} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Let $P$ be a general point on the line $l$, so $\overrightarrow{OP} = \begin{pmatrix} -2+4\lambda \\ 3+2\lambda \\ 1+\lambda \end{pmatrix}$

$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix} -2+4\lambda \\ 3+2\lambda \\ 1+\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -3+4\lambda \\ 2+2\lambda \\ 1+\lambda \end{pmatrix}$

$\left|\overrightarrow{CP}\right| = \sqrt{14} \Rightarrow (-3+4\lambda)^2 + (2+2\lambda)^2 + (1+\lambda)^2 = 14$

$\Rightarrow 9 - 24\lambda + 16\lambda^2 + 4 + 8\lambda + 4\lambda^2 + 1 + 2\lambda + \lambda^2 = 14$

$\Rightarrow 21\lambda^2 - 14\lambda = 0 \Rightarrow 3\lambda^2 - 2\lambda = 0 \Rightarrow (3\lambda - 2)\lambda = 0 \Rightarrow \lambda = 0$ or $\lambda = \frac{2}{3}$

When $\lambda = 0$, $\overrightarrow{OP} = (-2, 3, 1)$

When $\lambda = \frac{2}{3}$, $\overrightarrow{OP} = \left(\frac{2}{3}, \frac{13}{3}, \frac{5}{3}\right)$

So the circle intersects $l$ at $(-2, 3, 1)$ and $\left(\frac{2}{3}, \frac{13}{3}, \frac{5}{3}\right)$

**53)**

Show that $(40, 7, -20)$ lies in the plane:

$3(40) - 2(7) + 5(-20) = 120 - 14 - 100 = 6$ as required

Show that $\begin{pmatrix} -4 \\ 9 \\ 6 \end{pmatrix}$ is perpendicular to the plane:

$\begin{pmatrix} -4 \\ 9 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} = (-4) \times 3 + 9 \times (-2) + 6 \times 5$

$= -12 - 18 + 30 = 0$

So the line $l$ lies in the plane $\Pi$

**54)**

$\Pi: \mathbf{r} = \begin{pmatrix} 9 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 9 \\ -3 \\ -8 \end{pmatrix}$

**55)**

$$\begin{pmatrix}4\\6\\-3\end{pmatrix}\times\begin{pmatrix}2\\9\\18\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\4&6&-3\\2&9&18\end{vmatrix}$$

$$=(6\times18-(-3)\times9)\mathbf{i}-(4\times18-(-3)\times2)\mathbf{j}+(4\times9-6\times2)\mathbf{k}$$

$$=135\mathbf{i}-78\mathbf{j}+24\mathbf{k}$$

So the equation of the plane $\Pi$ is $135x-78y+24z=d$

Substitute in $(1,-6,7)$ to find $d$:

$135(1)-78(-6)+24(7)=d\Rightarrow d=771$

$\therefore 135x-78y+24z=771$

**56)**

$$\begin{pmatrix}2\\1\\9\end{pmatrix}\times\begin{pmatrix}-3\\5\\6\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&1&9\\-3&5&6\end{vmatrix}$$

$$=(1\times6-9\times5)\mathbf{i}-\big(2\times6-9\times(-3)\big)\mathbf{j}+\big(2\times5-1\times(-3)\big)\mathbf{k}$$

$$=-39\mathbf{i}-39\mathbf{j}+13\mathbf{k}$$

So the equation of the plane $\Pi$ is $-39x-39y+13z=d$

Substitute in $(2,6,-1)$ to find $d$:

$-39(2)-39(6)+13(-1)=d\Rightarrow d=-325$

$\therefore -39x-39y+13z=-325$

Substitute in $(4,k,-5)$ to find $k$:

$-39(4)-39(k)+13(-5)=-325$

$\Rightarrow -156-39k-65=-325$

$\Rightarrow 104-39k=0$

$\Rightarrow k=\frac{8}{3}$

**57)**

$$\begin{pmatrix}k\\3\\1\end{pmatrix}\times\begin{pmatrix}-1\\-2\\8\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\k&3&1\\-1&-2&8\end{vmatrix}$$

$=\big(3\times 8-1\times(-2)\big)\mathbf{i}-\big(k\times 8-1\times(-1)\big)\mathbf{j}+\big(k\times(-2)-3\times(-1)\big)\mathbf{k}$

$=26\mathbf{i}+(-8k-1)\mathbf{j}+(-2k+3)\mathbf{k}$

So the equation of the plane $\Pi$ is $26x+(-8k-1)y+(-2k+3)z=d$

Substitute in $(2,3,-8)$:

$26(2)+(-8k-1)(3)+(-2k+3)(-8)=d$

$\Rightarrow 52-24k-3+16k-24=d$

$\Rightarrow 25-8k=d$ (1)

Substitute in $(4,1,10)$:

$26(4)+(-8k-1)(1)+(-2k+3)(10)=d$

$\Rightarrow 104-8k-1-20k+30=d$

$\Rightarrow 133-28k=d$ (2)

Solve (1) and (2): $k=\frac{27}{5}$ and $d=-\frac{91}{5}$

So the equation of the plane is

$26x+\left(-8\left(\frac{27}{5}\right)-1\right)y+\left(-2\left(\frac{27}{5}\right)+3\right)z=-\frac{91}{5}$

$\Rightarrow 26x-\frac{221}{5}y-\frac{39}{5}z=-\frac{91}{5}$

$\Rightarrow 130x-221y-39z=-91$

**58) a)**

$\frac{x-0}{2}=\frac{y-0}{5}=\frac{z-0}{-7}$

or $\frac{x}{2}=\frac{y}{5}=\frac{z}{-7}$

**58) b)**

$\frac{x+2}{4}=\frac{y-3}{-1}=\frac{z-5}{1}$

or $\frac{x+2}{4}=3-y=z-5$

**58) c)**

$x=4,\frac{y+8}{1}=\frac{z-0}{6}$

or $x=4,y+8=\frac{z}{6}$

**58) d)**

$\frac{x-0}{3} = \frac{z-7}{-8}, y = 1$

or $\frac{x}{3} = \frac{7-z}{8}, y = 1$

**58) e)**

$\frac{x+9}{1} = \frac{y-3}{-5}, z = 8$

or $x + 9 = \frac{3-y}{5}, z = 8$

**58) f)**

$x + 2 = \lambda, y = 1, z = -6$

**59) a)**

$$\left(\mathbf{r} - \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}\right) \times \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \mathbf{0}$$

**59) b)**

$$\left(\mathbf{r} - \begin{pmatrix} 0 \\ -6 \\ 5 \end{pmatrix}\right) \times \begin{pmatrix} 4 \\ -2 \\ -5 \end{pmatrix} = \mathbf{0}$$